

# The Role of Mathematics in Machine Learning

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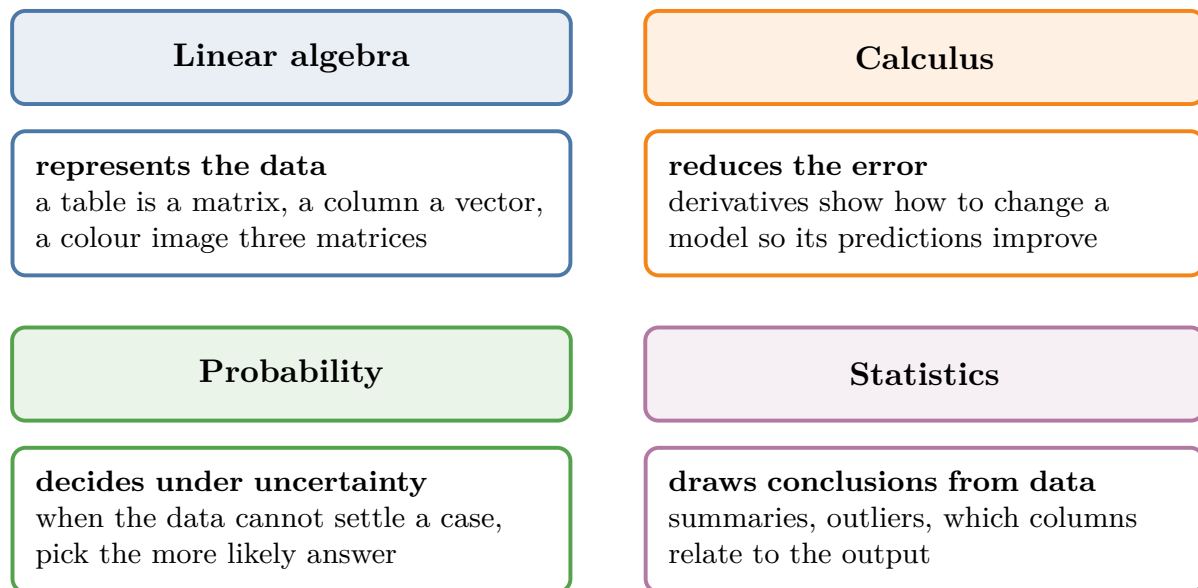
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## 1. Overview

**Key point:** Four branches of mathematics carry most of ML: linear algebra stores the data, calculus improves the model, probability decides when the data is unclear, and statistics draws conclusions from the data.

ML is not magic: every algorithm is mathematics running on data. To understand why an algorithm works, and to fix it when it does not, we need the mathematics underneath.

### Four branches of maths in ML



**Figure 1:** The four branches of mathematics used in ML, and the job each one does

Figure 1 shows the four branches and the job each one does. This Note gives a first feel for each job and points to the Notes where each branch is taught. The details come in those Notes.

## 2. Linear algebra: representing data

**Key point:** Linear algebra gives us the containers for data (vectors, matrices and tensors) and the operations that act on a whole container at once.

**Linear algebra** is the branch of mathematics that works with vectors and matrices. Its first job in ML is to store data in a form an algorithm can use:

- **A table** of numbers becomes a matrix: one row per example, one column per feature.
- **One column** becomes a vector.

- **A colour image** becomes three matrices of pixel values: one for red, one for green and one for blue.
- **A video** is a sequence of such images, so it becomes a stack of them.

These containers are tensors, explained in the [tensors Note](#).

Its second job is to act on a whole container in one step. “Multiply every row by 2” or “subtract the mean from every column” is a single operation on a matrix, not a loop over thousands of numbers. ML algorithms are written as such operations, which is why they can run on large data.

The [linear algebra roadmap Note](#) lists every linear algebra topic ML needs. The first of them, vectors, starts in the [vectors and feature vectors Note](#).

### 3. Calculus: reducing the error

**Key point:** Predictions are never perfect; calculus tells us how to change a model so that its error shrinks.

**Calculus** is the branch of mathematics about change: differentiation (slopes) and integration (areas). For beginners it is usually the hardest of the four.

An ML model predicts something, and its predictions are never exactly right. The gap between the prediction and the truth is the **error**. Training a model means changing it until the error is as small as we can get it, which is called **optimisation**.

Most optimisation methods are built on calculus. The derivative of the error tells us in which direction to change each part of the model so the error goes down. [Gradient descent](#) does exactly this, one small step at a time.

**Extra:** A model with 100% accuracy is a warning sign, not a success. Real data contains noise that no model can predict, so a perfect score usually means a mistake, such as test data leaking into training (data leakage, see the [toy project Note](#)).

Calculus first appears in the [linear regression maths Note](#), where a derivative set to zero gives the best line. Differential calculus gets its own treatment in a later maths Note.

### 4. Probability: deciding in the grey zone

**Key point:** When the data cannot settle a case for certain, probability tells us which answer is more likely, and we choose that one.

Often the data leaves a grey zone, where no answer is certain. Suppose we want to guess a person’s gender from their salary alone. In the salary range 4 to 5 lakh rupees, there are both men and women, so salary cannot decide.

Probability still lets us decide. If 40% of the people in that range are men and 60% are women, then for a new person earning 4.5 lakh rupees the more likely answer is “woman”, and that is what we predict. The chances inside one range add up to 100%.

This is how probabilistic classifiers such as [Naive Bayes](#) work: compute the probability of every class, then pick the largest. The rules of probability are taught from the [random experiments and events Note](#) to the [joint, marginal and conditional probability Note](#), and Bayes' theorem builds on them.

## 5. Statistics: drawing conclusions from data

**Key point:** Statistics is used the most of the four: from the moment we receive data to the moment we use the result.

Statistics is the branch of mathematics for collecting and analysing data so that we can draw conclusions from it (see the [what is statistics Note](#)). It appears at every stage of an ML project.

In real work, nobody hands us a dataset with a list of questions. We get the data and must find the meaningful questions ourselves, then answer them. Statistics supplies the tools:

- **Is there an outlier?** One value far from all the others can mislead a model (see the [outliers Note](#)).
- **Is the data noisy?** Noise decides how far we can trust the data.
- **Which algorithm suits this data?** The shape of the data points to suitable algorithms.
- **Is this column related to the output?** If not, there is no reason to feed it to the algorithm. Measures such as [correlation](#) answer this.

Data analysis, which ML depends on heavily, is built almost entirely on statistics (see the [understanding your data Note](#)). The [statistics roadmap Note](#) maps the whole subject.

## 6. Summary

| Branch         | Its job in ML                   | Example                          | Where it is taught                                          |
|----------------|---------------------------------|----------------------------------|-------------------------------------------------------------|
| Linear algebra | Represent data and act on it    | A colour image as three matrices | <a href="#">Roadmap Note</a> , <a href="#">tensors Note</a> |
| Calculus       | Reduce the error (optimisation) | Gradient descent                 | <a href="#">Gradient descent Note</a>                       |
| Probability    | Decide under uncertainty        | Pick the more likely class       | <a href="#">Events Note</a> onward                          |
| Statistics     | Draw conclusions from data      | Find outliers, related columns   | <a href="#">Roadmap Note</a>                                |

- ML is mathematics running on data; four branches carry most of it.
- Statistics is used the most; linear algebra does the heavy lifting of storing and transforming data.

## 7. Key terms

| Term           | Meaning                                                                 |
|----------------|-------------------------------------------------------------------------|
| Linear algebra | The branch of mathematics that works with vectors and matrices          |
| Calculus       | The branch of mathematics about change: differentiation and integration |
| Error          | The gap between a model's prediction and the true value                 |
| Optimisation   | Changing a model step by step until its error is as small as possible   |